

Sugarfit Technical Assignment: CGM Spike Predictor & Coaching Assistant

1. Overview

About Sugarfit

Sugarfit is a health-tech platform dedicated to helping users manage diabetes and improve metabolic health. By combining **Continuous Glucose Monitors (CGM)**, precise meal logging, and expert coaching, Sugarfit provides a data-driven approach to glucose management.

What is a CGM?

A Continuous Glucose Monitor (CGM) is a wearable sensor that tracks glucose levels in real-time, providing a constant stream of data points throughout the day. This allows users and coaches to see exactly how specific meals, activities, and stress levels impact metabolic health.

The Goal

Coaches and the product require an early warning system to identify when a user is likely to experience a glucose spike. This assignment tasks you with building an MVP that predicts these spikes and provides clear, actionable explanations grounded in Sugarfit's style.

2. Technical Requirements

Part A: Spike Prediction & Detailed Explanation

Objective: Predict whether a user's glucose will exceed a threshold in the next 2 hours (Binary Classification).

- **Data Sources:** You may use the OhioT1DM CGM Dataset, Tidepool CGM datasets, or create a synthetic dataset.
- **Modeling Requirements:** Train at least one baseline model, such as a logistic regression or a simple tree-based model.
- **Use a time-aware split for evaluation** to ensure no data leakage across time.
- **Feature Ideas:** Consider glucose trends (slope/delta) over the last 30–120 minutes, time since last meal, and carb counts.

Output Components:

- 1 **Probability of Spike:** The statistical likelihood (risk score) of exceeding the threshold.
- 2 **Detailed Explanation:** Provide a structured narrative explaining the factors leading to the risk (e.g., “The risk is high primarily due to a significant carb intake of 60g recorded 15 minutes ago, combined with a rising glucose velocity over the last half hour”).

Part B: Sample Dataset (Synthetic Example)

Candidates are not restricted to these values, but you may use the following structure as a starting point for synthetic data generation:

user_id	timestamp	glucose (mg/dL)	carbs (grams)	meal_type
USR_001	2023-10-01 08:00:00	95	0	N/A
USR_001	2023-10-01 08:15:00	98	45	Breakfast (Oats)

USR_001	2023-10-01 08:30:00	115	0	N/A
USR_001	2023-10-01 08:45:00	140	0	N/A

Part C: API Service

Objective: Build a small FastAPI (or similar) application to expose the system.

- Endpoint: POST /predict

Input Structure:

```
{ "user_id": "123", "recent_data": [...] }
```

Output Structure:

```
{ "will_spike": true, "risk_score": 0.78, "explanation": "Detailed narrative explaining the interplay between recent meal timing, carb load, and current glucose trend." }
```

3. Deliverables & Evaluation

Submission: A GitHub repository containing:

- README: Instructions on how to run the service, architecture overview, and design choices.
- Trade-offs: Documentation of what you skipped and why (focusing on an MVP approach).
- Testing: Sample curl or Postman calls to verify the API.

Evaluation Criteria:

- Data Thinking: Handling of time-series data and feature engineering.
- Engineering Clarity: Clean, understandable code and API design.
- Product Sense: The quality and logic of the detailed explanations provided for the risk.

Time Expectation: ~6–8 hours (1 day max). Focus on a clean, understandable MVP.